

12 Big Data Processing

▼ Introduction to Big Data



- Big Data refers to datasets that are too large or complex to be handled by traditional DBMS tools.

- Initially defined by **3 Vs**: Volume, Velocity, and Variety.
- Expanded to **5 Vs** with: Veracity and Value added by vendors and researchers.

▼ 5 Vs

▼ 1 Volume – "Data at Rest"



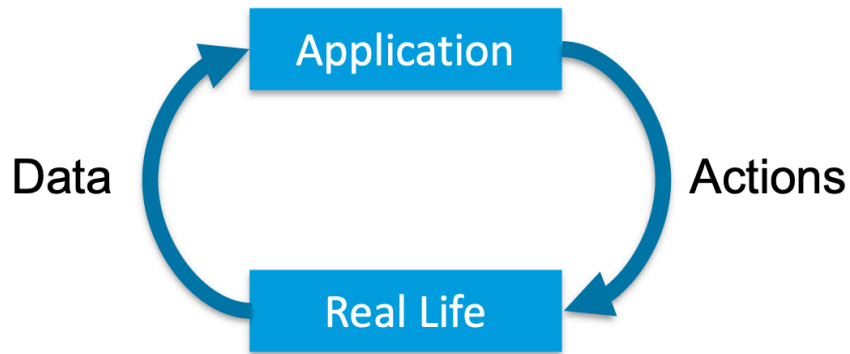
The **amount** of data generated, stored and processed.

- Refers to the vast amounts of data being generated.
- Sources include:
 - Enterprise systems (ERP, CRM, SCM).
 - Social networks (Facebook, Twitter).
 - Mobile phones and IoT devices.
- Challenge: **Scalable storage** and **distributed processing**.

▼ 2 Velocity – "Data in Motion"



The **speed** at which data is **generated, collected, and processed**.



- Example sources:
 - Streaming data (credit card swipes, video streams, phone calls).
- Key for enabling **real-time analytics** and competitive advantage.

▼ 3 Variety – "Data in Many Forms"



The **diversity of data types and sources**. Data can come in structured, semi-structured, or unstructured formats.

- Diversity in types and sources of data:
 - Structured: e.g. relational DBs.
 - Unstructured: images, tweets, videos, GPS logs, emails.
- Claim: Up to **80%** of data is unstructured.
- Requires multiple **data management** and **querying technologies**.

▼ 4 Veracity – "Data in Doubt"



The **accuracy** and **trustworthiness** of data.

- Data trustworthiness and quality.
- Challenges include:

- Inconsistencies and incompleteness.
- Ambiguity and latency.
- Need for validation depending on application criticality.

▼ 5 Value – "Data with Purpose"

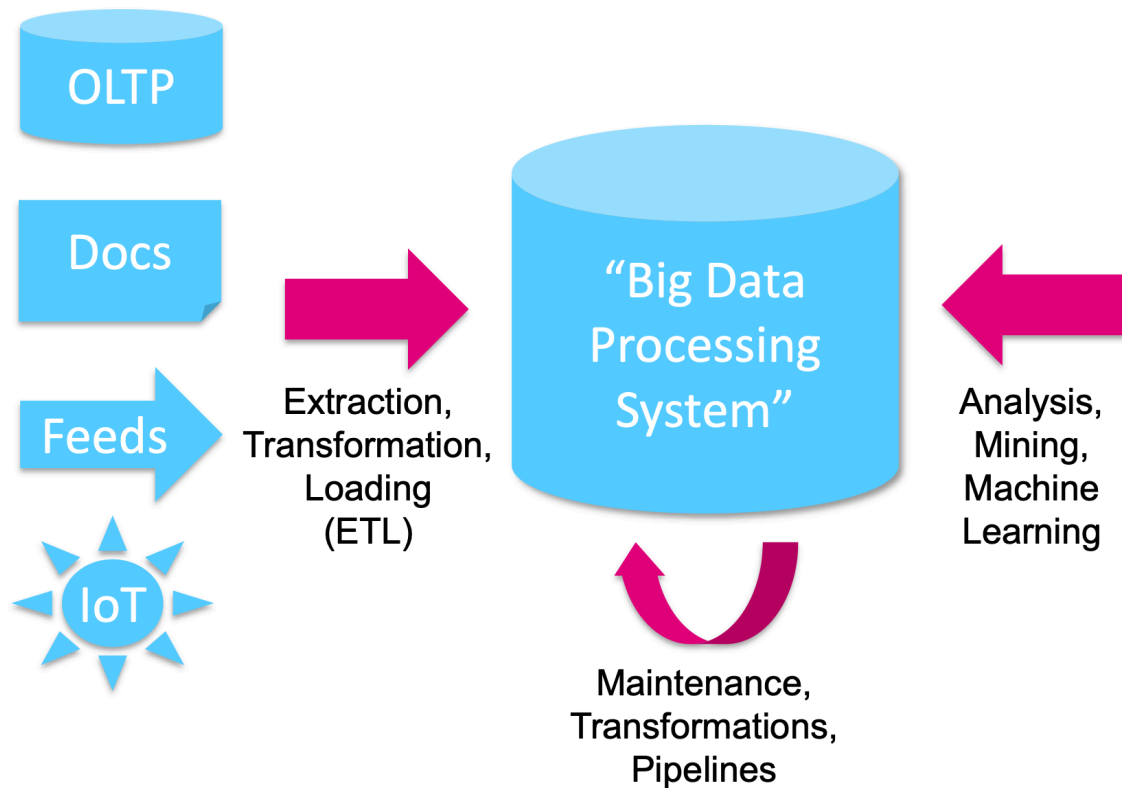


The ability to turn data into insights for decision **making and creating business value.**

- Refers to the **economic value** of Big Data.
- Concepts: ROI (Return on Investment), TCO (Total Cost of Ownership).
- Example: Microsoft's LinkedIn acquisition (\$260 per active user).
- Common pitfalls:
 - Jumping into Big Data without clear business goals.
 - Misjudging the need for Big Data infrastructure.

▼ Classes of Big Data Processing Systems

▼ Big Data Processing Overview



▼ Relational Database Systems

Strengths

- SQL programming API
- Huge SQL ecosystem
- ACID transactions
- Complex query processing

Weaknesses

- Little support for unstructured data
- Little support for machine learning training and serving

▼ MapReduce Systems

Strengths

- Distributed storage
- Great scalability

Weaknesses

- High latency, not suitable for real-time processing

- Able to process great variety of data sets (structured, semi-structured, unstructured)
- Great tool for distributed processing of large amounts of data
- Programming model can be complex and rigid
- Requires manual job orchestration and tuning
- Poor performance for small or iterative tasks
- Limited interactive data exploration capabilities

▼ Key-value Stores

Strengths

- Distributed storage
- Great scalability
- Short latency for single items
- Quick, "out of the box" way to store data
- Suitable for applications that involve frequent complex joins.

Weaknesses

- Limited storage model (only key, value pairs)
- Limited query interface (no SQL)
- No means to scan and filter data

▼ Document Databases

Strengths

- Store objects / XML / JSON in hierarchical form
- Good integration with object-oriented languages and JavaScript

Weaknesses

- Limited query interface (no SQL)
- No ACID guarantees
- Not designed for scans (very related to key-value stores)

▼ Graph Databases

Strengths

Weaknesses

- Capture graph relationships, e.g. knowledge graphs, social networks
- Fast at traversing edge chains, no joins needed
- Specific to graph applications
- Not many such use cases
- Relational databases outperform graph databases these days

▼ Data Lakehouses

Strengths

- Data storage in open data formats (Apache Parquet + Apache Iceberg) in the cloud
- Great ML support, e.g. training, but also Python-based notebooks

Weaknesses

- Typical API is Apache Spark, less versatile and supported than SQL
- However: Lakehouses are moving to SQL

Exercises